

FireKARE: Multimodal Biomechanical Risk Assessment for Incident Knee and Ankle Injuries in Firefighters

Pengyu Tao, Rui Xu, Jiamei Liang, Zhihao Qiao, Guanzheng Zhang, Guanjin Qu, and Huaming Wu

Abstract—Firefighters routinely perform physically demanding occupational tasks, including repeated landing, cutting, load carriage, and movement under unstable support conditions, which may increase the risk of knee and ankle injuries. Although Internet of Things (IoT) and multimodal sensing technologies enable detailed monitoring of firefighters’ movement states, transforming complex biomechanical signals into prospective injury-risk warnings remains challenging. This paper proposes FireKARE, a multimodal biomechanical risk assessment model for predicting incident knee and ankle injuries in firefighters. FireKARE constructs mechanism-aware biomarkers from electromyography (EMG), inertial measurement unit (IMU), ground reaction force (GRF), kinematic, kinetic, and muscle-synergy information collected during baseline movement tests, and employs a lightweight neural network to estimate one-year knee or ankle injury risk. We collected multimodal movement data from a real-world firefighter cohort and validated FireKARE using one-year follow-up outcomes. Experimental results demonstrate that FireKARE effectively predicts incident knee and ankle injury risk in firefighters, improving prediction accuracy by 7.5 percentage points compared with the best baseline. These findings suggest the potential of FireKARE for firefighter training-risk screening and individualized intervention support.

Index Terms—Deep learning, Biomechanics, firefighters, injury risk prediction, Internet of Things, knee injury, ankle injury.

I. INTRODUCTION

FIREFIGHTERS are routinely exposed to high-intensity physical activities during both training and emergency operations, such as loaded walking, jump landing, rapid cutting, stair climbing, and obstacle crossing. These movements impose substantial mechanical demands on the lower-limb joints under high-impact, high-load, and unstable-support conditions, thereby increasing the risk of musculoskeletal injuries among firefighters [1]. Among these injuries, knee and ankle disorders are of particular concern because they directly compromise mobility, load-bearing capacity, and operational readiness. Existing studies have investigated firefighter-related musculoskeletal injuries mainly from epidemiological, training-load, equipment-related, and fatigue-management perspectives

[2], [3]. Nevertheless, individualized and prospective early-warning models that leverage movement-test data to identify joint injury risks remain insufficiently explored.

Knee and ankle injury risk cannot be adequately characterized by a single physiological variable or joint-angle measurement. Sports-biomechanics studies have shown that lower-limb injury risk is jointly influenced by multiple biomechanical and neuromuscular factors, including neuromuscular control, ground reaction forces, knee-joint loading, hip control, and foot–ankle stability during landing and cutting tasks [4]–[6]. Meanwhile, systematic-review evidence suggests that knee-abduction kinematics or kinetics alone cannot consistently predict future anterior cruciate ligament injury [7]. These findings indicate that reliable injury-risk assessment should move beyond isolated indicators and jointly model multi-joint, multi-planar, and multimodal biomechanical features.

Internet of Things (IoT) and wearable sensing technologies have recently been applied to firefighter safety monitoring, training-state assessment, and early-warning applications. Existing studies have employed physiological and psychological signals to evaluate firefighters’ training states and injury-related conditions [8], while IoT-based early-warning systems and smart sensing architectures have also been developed for firefighting and industrial sensing scenarios [9], [10]. However, these studies primarily focus on real-time state monitoring, environmental hazard warning, or sensing-system design. They have rarely investigated the prospective association between baseline movement patterns and subsequent knee or ankle injury events. Therefore, a multimodal IoT-enabled framework for structural lower-limb injury warning in firefighters remains insufficiently explored.

To address this gap, this paper proposes a multimodal biomechanical risk assessment model for predicting incident knee and ankle injuries in firefighters. The proposed model transforms multisource sensing data collected from baseline movement tests into biomechanically meaningful risk representations and employs a lightweight deep-learning framework to estimate one-year injury risk. Different from studies centered on real-time physiological-state monitoring, psychological-state assessment, or action recognition, this work focuses on structural lower-limb injury risk and establishes an end-to-end pipeline from multimodal movement sensing to prospective injury warning. Validation on a real-world firefighter cohort with one-year follow-up data demonstrates that the proposed method can distinguish firefighters who subsequently develop knee or ankle injuries from those who remain uninjured, suggesting its potential for training-risk screening and individualized intervention support.

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II. MULTIMODAL BIOMECHANICAL INJURY WARNING MODEL

Although knee and ankle injury risk in firefighters is associated with multiple biomechanical factors, e.g., landing impact, joint loading, hip–knee–ankle control, foot–ankle stability, and neuromuscular coordination, prospective prediction from such heterogeneous signals remains challenging. Multimodal motion tests generate high-dimensional time-series data, whereas one-year follow-up observations usually contain only a limited number of injury-positive cases. Directly training high-capacity models on raw signals may therefore lead to overfitting, unstable decision boundaries, and limited interpretability.

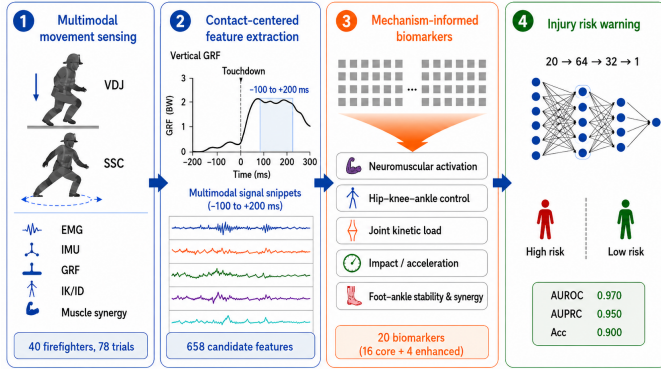


Fig. 1: Overall framework of the proposed FireKARE model for prospective multimodal biomechanical injury-risk warning in firefighters.

To address these challenges, this paper proposes FireKARE, a multimodal biomechanical injury-warning model for predicting incident knee and ankle injuries in firefighters. As illustrated in Fig. 1, FireKARE first transforms multisource baseline motion signals into low-dimensional, interpretable, and mechanism-aware biomarkers. It then employs a lightweight neural network to capture nonlinear risk patterns across these biomarkers. Probability calibration and threshold stabilization are further incorporated to produce individualized one-year injury-risk estimates.

A. Multimodal Motion Sensing and Biomarker Construction

Knee and ankle injury risk in firefighters arises from multiple interacting mechanisms, including landing impact, cutting support, hip–knee–ankle control, foot–ankle stability, and neuromuscular coordination. Therefore, FireKARE first converts multisource motion-sensing data into injury-mechanism-oriented biomarker representations, rather than directly modeling raw time-series signals.

Two baseline movement tasks were adopted: vertical drop jump (VDJ) and side-step cutting (SSC). VDJ mainly reflects vertical impact absorption and hip–knee control during landing, whereas SSC characterizes lateral support, rotational control, and foot–ankle stability during rapid changes of direction. These two tasks correspond to representative landing and cutting movements frequently involved in firefighter training and operational activities and jointly provide multidimensional information on knee and ankle injury risk.

Feature extraction was conducted within a contact-centered temporal window. For each trial, the initial contact was identified from the vertical ground reaction force (GRF), and a window spanning from 100 ms before to 200 ms after contact was extracted. Within this window, features were then derived from electromyography (EMG), inertial measurement unit (IMU), GRF, kinematic, kinetic, and muscle-synergy modalities, yielding 658 subject-level biomechanical descriptors.

To reduce overfitting under the limited cohort size, a domain-knowledge-guided strategy was used to compress these descriptors into a 20-dimensional biomarker vector. The selected biomarkers cover multiple sensing modalities and biomechanical mechanisms, including muscle activation timing, hip–knee–ankle motion control, joint loading, impact timing, foot–ankle stability, and muscle coordination. Specifically, the vector contains 16 primary features and four enhanced features, representing neuromuscular activation, hip and knee mechanics, movement stability, muscle synergy, task duration, GRF impact timing, mean knee moment, global acceleration peak, and mean hip rotation moment.

B. Biomarker-Based Injury Risk Assessment Model

After obtaining the 20-dimensional biomarker representation, FireKARE constructs a lightweight neural risk assessment model to estimate the probability of incident knee or ankle injury within one year. This module is designed to capture nonlinear risk interactions among multiple biomechanical biomarkers while maintaining low model capacity, stable decision behavior, and interpretability under limited-sample conditions. The design is motivated by the multifactorial nature of knee and ankle injury risk. In practical movement scenarios, injury outcomes are not determined by a single isolated indicator, but may result from the combined effects of landing-impact timing, knee-joint loading, hip adduction control, foot–ankle stability, and neuromuscular activation patterns. Therefore, FireKARE adopts a compact biomarker-level multilayer perceptron (MLP) to learn joint risk patterns in a low-dimensional mechanism-aware feature space. Compared with directly applying end-to-end temporal deep models to high-dimensional raw motion signals, this biomarker-based design is more suitable for prospective firefighter cohorts with limited sample sizes and imbalanced injury outcomes.

For each subject, the input to the model is the 20-dimensional biomarker vector, and the output is an injury-risk logit. The compact MLP controls model capacity by operating on mechanism-aware biomarker representations rather than raw multimodal time-series signals. This design allows the model to preserve nonlinear representation ability while reducing the risk of overfitting. Let $\mathbf{z}_i$ denote the biomarker vector of subject i , and let $f_\theta(\cdot)$ denote the compact MLP. To reduce prediction variance caused by random initialization, FireKARE first obtains an ensemble logit by averaging the outputs of S independently trained models:

$$p_i = \sigma(a f_\theta(\mathbf{z}_i) + b), \quad (1)$$

where $\mathbf{z}_i$ denotes the 20-dimensional biomarker vector, $f_\theta(\cdot)$ is the compact MLP that maps biomarkers to an injury-risk

logit, a and b are the Platt calibration parameters estimated from the training fold, and $\sigma(\cdot)$ is the sigmoid function. The resulting p_i represents the estimated one-year probability of incident knee or ankle injury.

To further improve the reliability of individualized warning decisions, FireKARE incorporates three small-sample robustification strategies. First, multi-seed logit ensembling reduces prediction variance induced by stochastic initialization. Second, Platt calibration maps ensemble logits into calibrated injury-risk probabilities, improving the interpretability of the predicted scores. Third, bootstrap threshold bagging estimates a stable classification threshold from out-of-fold probabilities in the training set, thereby reducing sensitivity to a single validation split or threshold selection. Together, these strategies enhance probability reliability and decision stability for small-sample prospective injury warning.

This module provides a lightweight, interpretable, and calibratable predictor for firefighter injury-risk screening. Beyond binary classification, it produces individualized risk probabilities that can support risk ranking, training-risk screening, and expert-assisted intervention planning.

C. Model Training and Validation

Each subject completed three valid trials for both the SSC and VDJ tasks, with 1–2 min rest intervals between trials to reduce fatigue effects. The SSC task involved a forward run followed by a single-leg landing and rapid lateral cutting to simulate deceleration and change-of-direction loading, whereas the VDJ task involved dropping from an elevated platform, landing on the force plate, immediately performing a vertical jump, and returning to stable standing. Trials were repeated when foot placement or movement execution did not satisfy the experimental protocol.

Model training and evaluation were conducted using subject-level five-fold cross-validation. In each fold, approximately 80% of the subjects were used for training and the remaining 20% were held out for testing, ensuring that no subject appeared in both sets. Missing-value imputation and feature standardization were fitted exclusively on the training subjects and then applied to the held-out test subjects to avoid information leakage.

During testing, FireKARE generated a positive-class probability for each held-out subject, representing the estimated one-year risk of incident knee or ankle injury. Performance was evaluated using AUROC, AUPRC, accuracy, sensitivity, specificity, and F1-score. This protocol evaluates the subject-level prospective warning ability of FireKARE under a strict train–test separation.

III. PERFORMANCE EVALUATION

A. Experimental Setup

To evaluate FireKARE, experiments were conducted using baseline motion-test data and one-year follow-up outcomes from 40 firefighters, including 11 injury-positive and 29 injury-negative subjects. After quality control, 78 valid motion-test records were retained and further aggregated into subject-level biomarker representations for model training and validation.

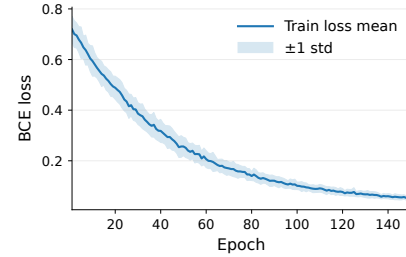
A subject-level five-fold cross-validation protocol was adopted to prevent trial-level information leakage. In each fold, all trials from the same subject were assigned exclusively to either the training set or the validation set. Missing-value imputation, feature standardization, probability calibration, and threshold selection were fitted using only the training subjects, while the held-out subjects were used solely for final evaluation.

FireKARE was implemented in Python using PyTorch. The model took the 20-dimensional mechanism-aware biomarker vector as input and produced a knee or ankle injury-risk logit. A lightweight MLP was combined with multi-seed logit ensembling, Platt calibration, and bootstrap threshold bagging to improve prediction stability and decision robustness under small-sample and class-imbalanced conditions.

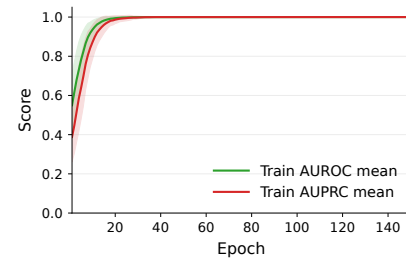
Four baseline methods were included for comparison: RBF-SVM, Random Forest, XGBoost, and a full-feature MLP trained on the 658-dimensional candidate feature set. Model performance was evaluated using AUROC, AUPRC, accuracy, sensitivity, specificity, and F1-score.

B. Model Convergence

To evaluate the training stability of FireKARE, we summarized the epoch-wise trajectories of BCE loss, AUROC, and AUPRC across five outer folds and ten random seeds per fold, resulting in 150 training runs.



(a) BCE loss curve during training



(b) Training AUROC and AUPRC trajectories

Fig. 2: Training convergence of FireKARE. Shaded regions denote ± 1 standard deviation across five outer folds and ten random seeds per fold.

As shown in Fig. 2a, the training BCE loss decreased steadily from 0.719 at epoch 1 to 0.052 at epoch 150, indicating stable optimization across folds and random seeds. Since the dataset is small and class-imbalanced, lower training loss does not necessarily lead to better generalization. Validation performance reached its best level around epoch 20 and showed no further improvement afterward; therefore, the epoch-20 checkpoint was selected for final testing.

TABLE I: Performance comparison of different models under subject-level five-fold validation.

Model	AUROC	AUPRC	Accuracy	Sensitivity	Specificity	F1
SVM-RBF	0.953 $\pm$ 0.043	0.917 $\pm$ 0.083	0.825 $\pm$ 0.068	0.733 $\pm$ 0.253	0.860 $\pm$ 0.080	0.687 $\pm$ 0.124
Random Forest	0.903 $\pm$ 0.110	0.800 $\pm$ 0.240	0.825 $\pm$ 0.112	0.467 $\pm$ 0.361	0.967 $\pm$ 0.075	0.533 $\pm$ 0.361
XGBoost	0.813 $\pm$ 0.162	0.674 $\pm$ 0.246	0.750 $\pm$ 0.088	0.100 $\pm$ 0.224	1.000 $\pm$ 0.000	0.133 $\pm$ 0.298
MLP-full features	0.310 $\pm$ 0.217	0.295 $\pm$ 0.089	0.725 $\pm$ 0.056	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000
FireKARE(ours)	0.970 $\pm$ 0.041	0.950 $\pm$ 0.075	0.900 $\pm$ 0.056	0.833 $\pm$ 0.236	0.933 $\pm$ 0.091	0.813 $\pm$ 0.119

Fig. 2b shows that training AUROC and AUPRC increased from 0.550 to 0.994 and from 0.388 to 0.987, respectively, suggesting that FireKARE progressively learned discriminative patterns from the biomarker representations. Overall, the convergence results indicate stable optimization, while validation-based checkpoint selection helps reduce overfitting in small-sample prospective injury-risk prediction.

C. Model Effectiveness

Table I compares FireKARE with baseline models under subject-level five-fold cross-validation. FireKARE achieved the best overall performance, obtaining the highest AUROC, AUPRC, accuracy, sensitivity, and F1-score. RBF-SVM showed competitive AUROC and AUPRC but lower sensitivity and F1-score, indicating less stable detection of injury-positive subjects. Random Forest, XGBoost, and the full-feature MLP exhibited negative-class bias, with XGBoost and the full-feature MLP achieving high specificity at the cost of low sensitivity and F1-score.

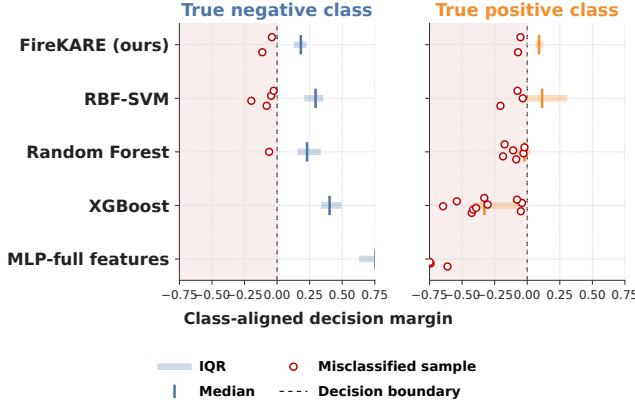


Fig. 3: Class-aligned decision margins across models. Positive and negative margins indicate correct classifications and misclassifications, respectively. Horizontal bars represent the interquartile range, vertical lines indicate the median, and red hollow circles denote misclassified samples.

To further examine classification stability, we calculated the class-aligned decision margin. For true-negative samples, the margin was defined as the classification threshold minus the predicted positive probability; for true-positive samples, it was defined as the predicted positive probability minus the classification threshold. Positive margins indicate correct classification, whereas negative margins indicate misclassification. As shown in Fig. 3, FireKARE produced consistently positive margins for both injury-negative and injury-positive subjects. By contrast, the positive-sample margins of Random Forest, XGBoost, and the full-feature MLP were largely located on the misclassified side, indicating a tendency toward false-negative predictions.

Overall, FireKARE provided more balanced discrimination between injury-positive and injury-negative firefighters. Its advantage is reflected not only in ranking and classification metrics, but also in reducing the missed detection of high-risk subjects while maintaining a low false-alarm level. These results support the effectiveness of mechanism-aware biomarkers, lightweight neural modeling, probability calibration, and threshold stabilization for small-sample subject-level injury prediction.

IV. CONCLUSION

This paper proposed FireKARE, a multimodal biomechanical model for one-year knee and ankle injury warning in firefighters. By converting baseline motion data into mechanism-aware biomarkers, FireKARE achieved stable convergence and outperformed baseline models, with an AUROC of 0.970, an AUPRC of 0.950, and an accuracy of 0.900. Future work will validate the model on larger external cohorts and incorporate factors such as training load, injury history, and fatigue status.

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